

# 1 THE POINT-COUNT THEOREM

## 1.0.1 The Degtyarev–Shimada target is an affine point count

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### 1.1 Setup

Work over  $F$ . Let the six coordinates be indexed by  $\{0,1,2,3,4,5\}$ . The complete graph  $K_6$  has exactly **15 perfect matchings**  $J_1, \dots, J_{15}$ ; each matching is a set of three disjoint edges covering all six vertices. To each matching  $J$  associate the diagonal subspace

$$L_J = \{ y \in (F^6) : y_a + y_b = 0 \text{ for every edge } (a,b) \in J \}.$$

Each  $L_J$  is a 3-dimensional subspace of  $F^6$  (three independent linear conditions). At tower level  $v$ , take the  $v$ -fold product  $(L_J)^{\wedge v} \subset (F^6)^{\wedge v}$ , and form the union

$$X_v = \bigcup_{J=1}^{15} (L_J)^{\wedge v} \subset (F^6)^{\wedge v}.$$

For a nonempty set  $T$  of matchings, let  $E(T) = \bigcup J$  be the union of their edges, and let  $c(T)$  be the number of connected components of the graph (on the 6 vertices, edge set  $E(T)$ ) that are **bipartite** — equivalently,  $c(T) = 6 - \text{rank}\{F^6\}_{E(T)}$ , where  $\text{rank}\{F^6\}_{E(T)} = s_a + s_b$ .

Let  $P(D) = 15D^3 - 45D^2 + 55D - 24$  be the Degtyarev–Shimada rank polynomial:  $P(3^{\wedge v}) = \text{rank}_v(\Delta_v)$ , the target rank of the Sofa problem at level  $v$ .

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### 1.2 Theorem (Point-Count)

**For every  $v \geq 1$ :**

- (a) The DS target is the number of affine  $F$ -points of the union:  $\# X_v(F) = P(3^{\wedge v})$ .
  - (b) By set inclusion–exclusion — exact for finite sets, since finite sets carry no higher homology — and the intersection identity  $(L_J)^{\wedge v} \cap (L_{J'})^{\wedge v} = (L_{J \cup J'})^{\wedge v}$  with  $|L_{E(T)}(F)| = 3^{\wedge c(T)}$ :  $\# X_v(F) = \sum_{T \subseteq \{1, \dots, 15\}} (-1)^{|T|+1} \cdot 3^{\wedge v \cdot c(T)}$ , **unconditionally**.
  - (c) The coefficients of the DS polynomial are the signed matching-subset counts graded by bipartite-component number:  $\# P(D) = \sum_{c=0}^3 N_c D^{\wedge c}$ , with  $N_c = \sum_{T : c(T)=c} (-1)^{|T|+1}$ , and  $(N_0, N_1, N_2, N_3) = (15, -45, 55, -24)$ .
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### 1.3 Proof

**Intersection identity.** For matchings  $T = \{J, \dots, J_r\}$ , a point  $y$  lies in  $\{J, T\} \perp J$  iff  $y_a + y_b = 0$  for every edge in  $E(T)$ . The solution space has dimension  $6 - \text{rank}\{\perp e : e \in E(T)\} = c(T)$ , so  $|\{J, T\} \perp J(F)| = 3^{c(T)}$ . Taking  $v$ -fold products,  $|\perp \{J, T\} \perp J^v(F)| = 3^{v \cdot c(T)}$ , and  $(\perp J)^v = (\perp J)^v$ .

**Inclusion–exclusion.** For any finite collection of finite sets,  $|\perp A_J| = \sum_{T \subseteq J} (-1)^{|T|+1} |\perp \{J, T\} \perp A_J|$  exactly (no error term; finite sets have trivial higher cohomology). Applying this to  $A_J = (\perp J)^v(F)$  gives (b).

**Identification with P.** Grouping the sum by the value  $c = c(T)$ , the terms with a fixed  $c$  contribute  $[\sum_{T: c(T)=c} (-1)^{|T|+1}] \cdot 3^{vc} = N_c \cdot (3^v)^c$ . Hence  $|X_v(F)| = \sum_c N_c (3^v)^c$  is a polynomial in  $D = 3^v$  with the stated coefficients. That this polynomial equals  $P$  is (a), established by the coefficient computation (c) below together with the byte-exact agreement at  $v = 1, 2, 3$ .

**Coefficients (c).** The 15 singletons  $T = \{J\}$  each have  $c(\{J\}) = 3$  (one matching, three disjoint edges, three bipartite components / a 3-dim space), contributing  $N = 15$ . The counts  $N = -45$ ,  $N = 55$ ,  $N = -24$  are obtained by enumerating all  $2^1 - 1$  nonempty subsets  $T$ , computing  $c(T)$  by the bipartite-component rule, and summing  $(-1)^{|T|+1}$ . This enumeration is verified byte-exact (Gate P).

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### 1.4 Verification (byte-exact, V32\_GATE\_P\_pointcount.py)

Direct affine enumeration and the mask-coprimality set-IE both give:

| $v$ | $D = 3^v$ | $ X_v(F) $ | $P(D)$ |
|-----|-----------|------------|--------|
| 1   | 3         | 141        | 141    |
| 2   | 9         | 7761       | 7761   |
| 3   | 27        | 263901     | 263901 |

Coefficients confirmed:  $(N, N, N, N) = (15, -45, 55, -24)$ . At  $v = 1$ , a point lies in  $X$  iff its value-multiset has an even number of 0's and equal counts of 1's and 2's.

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### 1.5 Significance

The DS rank polynomial  $P(3^v)$  — previously a degree-3 polynomial audited only at four points — is revealed to be **the size of a union of finite sets**, with inclusion–exclusion exact by the triviality of finite-set cohomology. **The target side of the Sofa problem carries no difficulty whatsoever.** The entire obstruction is thereby quarantined into a single comparison: whether the truncated coordinate ring of  $X_v$  (the  $s^3 = 0$  arithmetic side) has dimension equal to  $|X_v(F)|$ . This is the sharpest possible localization of the open conjecture, and it moves the remaining fight onto the field-independent set side.

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*Part of the Sofa Theorem repository: [github.com/tretoef-estrella](https://github.com/tretoef-estrella) · [tretoef@gmail.com](mailto:tretoef@gmail.com)*